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Symbols

⧫	*concept* — a term and its definition
⊙	*claim* — a single truth-apt proposition
⊢	*argument* — premises leading to a conclusion
★	*position* — a named standpoint or view
?	*question* — an open issue under debate

List of Entries

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A

⊢ Ability hypothesis

The Ability Hypothesis — experience teaches know-how, so Mary learns no new fact

Argument · after Lewis

The *Ability Hypothesis* is David Lewis's reply to the Knowledge Argument, also defended by Laurence Nemirow. It holds that knowing what an experience is like is not a piece of factual knowledge at all, but a bundle of *abilities*: to recognise the experience, to imagine it, and to remember it. If that is right, then when Mary the colour scientist, who knows every physical fact about colour vision but has never seen red, finally sees it, she gains know-how rather than a new fact, and the inference from her learning something to the falsity of physicalism is blocked.

The argument

The attack lands on the Knowledge Argument's verdict step: the critical question (CQ2 of the thought-experiment scheme) of how to *read* the agreed intuition that Mary learns something. The intuition is robust, but reading it as *acquiring a new propositional fact* is theory-laden, and the verdict survives under a deflationary reading that the original argument cannot use. Given an analysis of "knowing what it is like", the inference is deductive:

1. Knowing what an experience is like consists in possessing certain abilities: to recognise the experience when it occurs, to imagine it, and to remember it (Lewis 1988; Nemirow 1990).
2. On first seeing red, what Mary gains is exactly this: knowing what seeing red is like.
3. Abilities are not propositional knowledge; gaining one is acquiring a skill, not learning a truth.

∴ Mary gains no new fact. So the step from her epistemic progress to truths about experience that no physical description yields is blocked: there is no *truth* she lacked.

What it concedes

The move is notable for conceding both premises of the Knowledge Argument as written. It grants that Mary knew all the physical facts outright, and that she learns something on release under the know-how reading. What it severs is the inference between them. "Mary learns something" is true; "Mary learns a new fact" is what the conclusion needs; and the case cannot fund the stronger reading without begging the question against physicalism.

The weak point

The most disputed premise is the first, the ability analysis itself. Three counters are standard. Mary seems able to *use* her new knowledge in reasoning and inference ("seeing orange must be like *this*, only more yellowish"), which bare abilities do not explain. One can apparently have the abilities without the knowledge (a distracted imaginer who can summon red but is not thereby informed), and arguably the knowledge while lacking the imaginative ability (someone paralysed of imagination). And the acquaintance theorist offers a rival non-factual analysis, that what Mary gains is acquaintance rather than a fact, so even granting that no new fact is learned, the ability analysis is not the only option.

> This is the most famous escape from the Mary story (the Knowledge Argument: since Mary, who knew all the physics of colour, still learns something when she first sees red, physics must be incomplete). The escape turns on an ambiguity in "learning". Learning a *fact* (that Paris is the capital of France) is one thing; picking up a *skill* (juggling) is another. On this view Mary picks up skills, namely recognising, imagining, and remembering red. No fact was missing from her books, so nothing was missing from physics. The catch: her new grasp of red seems usable in actual reasoning, the way facts are and skills are not, so perhaps it is not *just* a skill.

See also

- ▹ Knowledge argument (Mary's room) — the argument this one attacks: that Mary learns a fact the physical description omits.
- ▹ Knowledge argument — the Mary thought experiment itself.
- ▹ Mary gains abilities — the conclusion: Mary acquires abilities, not a new fact.
- ▹ Mary gains acquaintance — the rival non-

factual reply: acquaintance rather than know-how.

- ▷ NON-DEDUCIBLE EXPERIENTIAL TRUTHS — the anti-physicalist conclusion this attack is meant to block.
- ▷ LEWIS — WHAT EXPERIENCE TEACHES (1988) — Lewis's "What Experience Teaches" (1988), the primary statement.
- ▷ NEMIROW — PHYSICALISM AND THE COGNITIVE ROLE OF ACQUAINTANCE (1990) — Nemirow's independent defence of the ability analysis.

⊙ Absence of an experiential threshold

There is no non-arbitrary point in nature at which experience first appears

Claim · after Nagel

If experience were confined to brains, or to life, or to systems above some threshold of complexity, there ought to be a principled place where it first switches on — a line such that below it there is nothing it is like to be the system and above it there is. No such non-arbitrary line is in view. The relevant differences across any proposed boundary look like differences of degree and organisation, not the kind of difference that could mark the abrupt entry of an entirely new ontological feature into the world.

This is the premise of the CONTINUITY ARGUMENT for panpsychism: absent a principled threshold, the parsimonious option is that experience, in some attenuated form, reaches all the way down. It is denied by emergentists and physicalists who hold there *is* a principled threshold — a level of integrated information or of functional organisation at which experience genuinely, if strongly, emerges (see ACCEPTABILITY OF STRONG EMERGENCE). It differs from NO RADICAL EMERGENCE: that claim says experience could not brutely *appear* at all from the non-experiential, whereas this one says that even granting it could, there is no non-arbitrary *place* to locate its first appearance.

SEE ALSO

- ▷ CONTINUITY ARGUMENT — the argument this premise drives.
- ▷ ACCEPTABILITY OF STRONG EMERGENCE — its denial: there *is* a principled emergence threshold.
- ▷ NO RADICAL EMERGENCE — the neighbouring premise about *whether*, not *where*, experience could emerge.

⊙ Acceptability of strong emergence

Strong emergence of experience from non-experiential matter is acceptable

Claim

Experience can arise as a genuinely novel, fundamental feature at some level of physical organisation — a brain, say — without being present, even in proto-form, in that organisation's parts. On this view *strong* emergence (the appearance of properties not grounded in the powers of the base) is a coherent and even expected feature of a layered natural world, not the magic its critics allege. The dispute is specifically about strong emergence of the *phenomenal*; *weak* emergence, where higher-level patterns are fully grounded in and predictable from the base, is accepted by almost everyone and is not at issue here.

This is the standard physicalist denial of NO RADICAL EMERGENCE, and so the load-bearing premise of the EMERGENCE REPLY that undercuts the genetic argument for panpsychism: if strong emergence is acceptable, the genetic argument's choice between fundamental experience and unintelligible magic is a false dilemma. It is denied by panpsychists, who hold that strong emergence of the phenomenal is unintelligible (Strawson), and by those who think calling experience "strongly emergent" merely relabels the explanatory gap rather than closing it.

SEE ALSO

- ▷ NO RADICAL EMERGENCE — the claim this one denies.
- ▷ EMERGENCE REPLY — the objection this premise drives against the genetic argument.
- ▷ GENETIC ARGUMENT — the argument that the acceptability of strong emergence is meant to undercut.

♦ Access consciousness

Concept · after Block

Access consciousness is the functional sense of "conscious": a mental state is access-conscious when its content is *available* to the rest of the mind, poised for use in reasoning, for the direct rational control of action, and for verbal report. The term is Ned Block's (he abbreviates it *A-consciousness*). It is a matter of what a system can *do with* a state's content, settled entirely by the state's role in the cognitive economy and not at all by how, or whether, it feels.

A simple test fixes the idea: when you read this sentence, its meaning is poised for you to report, question, and reason with, and that availability is the sentence's access-consciousness. Theories of consciousness in the *global workspace* family (see GLOBAL WORKSPACE THEORY) effectively identify being conscious with being access-conscious in this sense, broadcast widely enough across the system to be used.

Access consciousness is defined by contrast with PHENOMENAL CONSCIOUSNESS, the felt or "what-it-is-like" side of experience. Block's central claim is that the two can come apart in both directions: a system might have the availability with no accompanying feel, and experience might even *overflow* what gets globally broadcast. That dissociation is set out at ACCESS VS. PHENOMENAL CONSCIOUSNESS. Keeping the two apart is the standard guard against treating a system's fluent self-report, an access-style capacity, as evidence that it *feels* anything.

> Calling a thought "access-conscious" just means the information is *available* to the rest of your mind, so that you can act on it, talk about it, and reason with it. It says nothing about whether the thought *feels* like anything. That is the whole point of the label: it lets us separate "the system can use and report this" from "there's something it's like to have it", so we don't slide from a machine describing its own inner states to assuming it must therefore feel them.

SEE ALSO

- PHENOMENAL CONSCIOUSNESS — the felt sense of consciousness it is contrasted with.
- ACCESS VS. PHENOMENAL CONSCIOUSNESS — Block's distinction and the case that the two dissociate in both directions.
- GLOBAL WORKSPACE THEORY — a theory that ties consciousness to access (global broadcast).
- FIRST-PERSON ACCESS — privileged first-person access, a distinct, epistemic notion.

♦ Access vs. phenomenal consciousness

Access consciousness vs. phenomenal consciousness

Concept · after Block

Access consciousness and *phenomenal consciousness* are two distinct things the single word "consciousness" can mean, separated by Ned Block ("On a Confusion about a Function of Consciousness", 1995). A state is *access-conscious* when its content is available to the rest of the mind — poised for use in reasoning, for the direct rational control of action, and for verbal report. A state is *phenomenally conscious* when there is something it is like to be in it — when it carries a felt, qualitative character. The first is a matter of what a system can *do with* a state's content; the second, of how the state *feels* from the inside.

THE TWO SENSES

ACCESS CONSCIOUSNESS (*A-consciousness*) is a functional, information-availability notion: a content is A-conscious to the degree that it is *poised* to be drawn on by the systems that reason, decide, and speak. Whether a state qualifies is settled entirely by the role its content plays in the cognitive economy — what downstream processes can reach and use it — and by nothing about how it feels. When you read this sentence, its meaning is poised for report and inference; that availability is its access side.

Phenomenal consciousness (P-consciousness) is the "what-it-is-like" aspect of a mental state: the redness of red, the hurt of a pain, the timbre of a heard note. It is the felt or qualitative character of experience, and it is the property at issue in the HARD PROBLEM of consciousness — the question of why any physical process should be accompanied by felt experience at all. The particular look of these black marks on the page, over and above your ability to report and reason about the words, is their phenomenal side.

WHY THE DISTINCTION MATTERS

The point of drawing the line is that the two can come apart, at least in principle, in *both* directions. A system might carry richly reportable models of its own inner states — answering fluently about what it is "noticing" or "feeling" — while there is nothing it is like to be that system at all: access without any accompanying feel. Block argues the converse is possible too, that phenomenal consciousness may *overflow* access, so that experience holds more than ever gets globally broadcast for report. The canonical illustration is a brief flash of a full array of letters: one seems to see the whole array vividly, yet can read out only a few before the image fades — as though the felt richness outruns what access captures. So "has A" neither entails nor is entailed by "has P."

This independence is the pivot on which several moves in the debate over machine introspection turn. It marks the central danger: mistaking a system's *functional, access-style* capacity — its ability to report on its own states, as some current language models do to a limited and unreliable degree (see LLM FUNCTIONAL INTROSPECtion) — for evidence of a *phenomenal* inner life. It also yields a subtler observation: *posing* the hard problem from the first-person side is itself an access-dependent act, requiring access to a representation of one's own phenomenal-seeming states (see Posing the gap requires access), even though merely *having* phenomenal states requires no such access.

> "Conscious" runs together two different things. One is **access**: the information is available to the rest of your mind — you can reason with it, act on it, and say it out loud. The other is **feel**: there's something it's like to undergo the state (the redness of red, the hurt of pain). They're not the same. You could imagine a machine that has all the *access* — it reports its inner states fluently — with no *feel* at all; and some philosophers think the reverse can happen too, that we feel more than we can ever report. Keeping the two apart is the main guard against a tempting mistake: seeing a system talk about its own insides and concluding it must therefore *feel* something.

See also

- Access consciousness — the functional half of the distinction, treated on its own.
- Phenomenal consciousness — the felt half of the distinction, treated on its own.
- Hard problem — the felt, phenomenal side is exactly what this problem is about; access is precisely what it is *not*.
- LLM functional introspection — a candidate case of access-style self-report in machines, and the prime spot where it must not be read as the phenomenal kind.
- Posing the gap requires access — that stating the hard problem is itself an access-dependent act, even though undergoing phenomenal states is not.

C

Categorical, not dispositional

Phenomenal consciousness is categorical, not merely dispositional

Claim

Phenomenal consciousness is an *occurrent, categorical* matter — it is actually, presently *there* — rather than a merely *dispositional* affair, a matter of what a system *would* do under inputs it may never actually receive. A disposition is an "iffy" property: glass counts as fragile in virtue of how it *would* behave if struck, even if it is never struck. Experience does not look like that. It looks like the paradigm of a categorical fact, something occurrently happening, not a standing counterfactual profile.

The standard support for the contrast is that dispositions require a *categorical basis* — fragility holds in virtue of some actual microstructure of the glass — a demand associated with D. M. Armstrong (see ARMSTRONG — A MATERIALIST THEORY OF THE MIND (1968)); on the Russellian side, the intrinsic, categorical nature underlying structure is what Galen Strawson (see STRAWSON — REALISTIC MONISM: WHY PHYSICALISM ENTAILS PANPSYCHISM (2006)) presses as the home of experience. The claim itself stays modest about *what* the categorical basis is: it does not say whether that basis is the system's current physical configuration or some intrinsic nature beneath it. It asserts only that the phenomenal is categorical, not dispositional.

A dispositionalist or functionalist about consciousness would deny it, holding that to be in a phenomenal state just *is* to occupy a causal-dispositional role. The claim is kin to, but distinct from, PHENOMENAL DETERMINACY: determinacy says there is a *definite* fact about one's current phenomenal state, whereas this says the fact is *categorical/occurrent* rather than dispositional. The two function as companion premises behind THE DECOMPOSITION-INDETERMINACY OBJECTION and THE STRUCTURE-AND-DYNAMICS ARGUMENT respectively.

> A "disposition" is an iffy, would-do fact: saying glass is fragile means it *would* break if struck, even if it never actually is. A "categorical" fact is how something actually, presently *is*. This claim says your conscious experience is the second kind — it's really there right now, not just a bundle of "what the system would do if poked." A pain isn't a standing tendency to react; it's an occurrent felt thing happening. (Anyone who defines a mental state as nothing but a causal role would push back.)

SEE ALSO

- PHENOMENAL DETERMINACY — the companion claim that there is a *definite* fact about one's current experience; this one adds that the fact is *categorical*.
- DECOMPOSITION-INDETERMINACY OBJECTION — relies on determinacy; pairs with this claim's categoricity in the case against role-based consciousness.
- STRUCTURE-AND-DYNAMICS ARGUMENT — the argument this claim feeds: function captures only structure and dynamics, never the categorical/intrinsic.

Chinese nation

Chinese Nation (homunculi-headed robot)

Concept · after Block

The *Chinese Nation* is a thought experiment devised by Ned Block in "Troubles with Functionalism" (1978). Imagine the citizens of a whole country, or a head full of little people, given radios and a rulebook so that together they realise the entire functional and causal organisation of a single human mind: the same state-to-state transition structure a brain runs through, for an hour or so. By construction the system has exactly the inner organisation a conscious brain has. The question Block presses is whether it would therefore be conscious. Would there be something it is like to be the nation, or is there simply nobody home?

WHAT MAKES IT DISTINCTIVE

The case has its bite precisely because it instantiates the functional organisation, not merely the profile of inputs and outputs a mind produces. It is therefore not the "right behaviour, no processing" case of Block's other machine, THE LOOKUP-TABLE MIND, but the "right processing, yet intuitively nobody home" case. That is what makes it tell against FUNCTIONALISM proper rather than against mere behaviourism: it grants everything a functionalist asks for and still seems to lack experience.

The dispute it feeds

The Chinese Nation is pressed on whether realising a mind's functional organisation is *sufficient for PHENOMENAL CONSCIOUSNESS*. The intuition that the nation has none drives THE CLAIM THAT SUCH A POPULATION WOULD FEEL NOTHING and the argument THAT ORGANISATION IS NOT SUFFICIENT FOR CONSCIOUSNESS, which in turn support the PHENOMENAL RESIDUE view and attack the uniform reading of SUBSTRATE INDEPENDENCE.

Replies

Two replies are characteristic, and both differ from the ones the lookup table invites. The *systems reply* denies the intuition outright: the nation taken as a whole may be a subject of experience even though no individual citizen is (Lycan, Dennett). A second worry concerns *speed and timing*, whether so slow and distributed a system genuinely realises the organisation at all. There is also a pointed structural observation. ORGANISATIONAL-ROLE FUNCTIONALISM cannot cheaply exclude the Chinese Nation the way it excludes the lookup table, because here the organisation really is present; that is exactly why the case is sharpened into THE DENSE CHINA-BRAIN REGRESS against the organisational grain.

A literary illustration

A vivid cousin of the case appears in Liu Cixin's science-fiction novel *The Three-Body Problem* (see LIU CIXIN). There the First Emperor Qin Shi Huang, prompted by von Neumann, ranks thirty million soldiers across a plain, each small unit raising black-and-white flags while cavalry gallop the signals between formations, to assemble a vast human machine. In the novel the machine is a computer crunching the orbits of three suns; re-task it to run a *brain* instead. Let each cluster of soldiers be a neuron, each raised flag its firing, the messengers its axons, and the whole drilled host step for one hour through exactly the state-to-state transitions a human cortex would. The army is now no calculator but the Chinese Nation itself: a country physically realising a mind's causal organisation while no single soldier feels a thing. If the soldiers enact the right transitions, is someone *over and above* them undergoing the experience, or is there nobody home?

> Imagine the entire population of a country handed radios and a rulebook and told to each act as one brain cell: when you receive these signals, send those. Follow the rules fast enough and the nation as a whole mimics, step for step, everything one human brain does inside, not just the outward behaviour but the actual inner wiring. Now the unsettling question: the country is doing literally everything your brain does when you feel pain, so is the *country* feeling pain? Is there one extra "someone", the nation itself, having an experience, hovering above millions of people who each feel nothing out of the ordinary? If running the right pattern is all it takes to be conscious, you seem pushed to say yes, yet most people's gut says surely not. That clash is the whole point of the thought experiment. It is a sharper challenge than a dumb answer-lookup machine, because here the full inner organisation really is present, so you cannot dismiss it for having no inner structure.

See also

- ▷ LOOKUP-TABLE MIND — Block's other machine: right behaviour with no inner processing, the case this one is defined against.
- ▷ FUNCTIONALISM — the theory the thought experiment targets: mind as functional organisation.
- ▷ CHINESE NATION (ABSENT QUALIA) — the argument that the realised organisation still lacks qualia, so organisation is not sufficient for consciousness.
- ▷ PHENOMENAL RESIDUE — the view the absent-qualia intuition supports: experience outruns functional role.
- ▷ SUBSTRATE-INDEPENDENCE THESIS — the thesis whose uniform reading the case attacks.
- ▷ DENSE CHINA-BRAIN REGRESS — where the case is sharpened against organisational-role functionalism.
- ▷ PHENOMENAL CONSCIOUSNESS — what the nation is alleged to lack despite the right organisation.
- ▷ BLOCK — TROUBLES WITH FUNCTIONALISM (1978) — Ned Block's "Troubles with Functionalism" (1978), where the thought experiment is introduced.

⧫ Chinese room

The Chinese Room

Concept · *after Searle*

The *Chinese Room* is John Searle's thought experiment against the claim that running the right computer program is, by itself, enough to produce a mind (see SEARLE — MINDS, BRAINS, AND PROGRAMS (1980)). A monolingual English speaker is sealed in a room with a rulebook — in effect, a program — that tells him how to manipulate Chinese symbols purely by their shapes. Chinese questions are slipped in; by following the rules he passes back Chinese answers indistinguishable from a native speaker's. Yet he understands no Chinese. The room, Searle argues, has the right *syntax* (formal symbol manipulation) without any *semantics* (meaning, understanding, intentionality) — and a digital computer, which does nothing but manipulate symbols by their shapes, is in exactly the same position.

WHAT IT TARGETS

The argument is aimed at **strong AI**, the computational-functionalist thesis that *implementing the right program is sufficient for a mind.* That precise target is what separates it from two neighbouring thought experiments with which it is often confused. Ned Block's LOOKUP-TABLE MIND ("Blockhead") attacks the sufficiency of *behaviour*: it produces the right outputs with **no** internal processing at all, just a giant table of canned answers. The Chinese Room, by contrast, *does* run a genuine program; what it denies is that *computation* suffices for *understanding*. Block's CHINESE NATION differs in the other direction: there a population realises a mind's full functional organisation, and the case is pressed against *qualia* — felt experience — whereas the Room is pressed against *intentionality*, whether the symbols mean anything to the system at all. Three different experiments, three different sufficiency theses.

CHARACTERISTIC REPLIES

Each of the standard replies is an objection-in-waiting. The **Systems Reply** grants that the man does not understand Chinese but locates the understanding in the whole system — man, rulebook, and room together — much as no single neuron understands English. The **Robot Reply** embeds the program in a robot body with cameras and limbs, so that its symbols stand in causal commerce with the world and thereby acquire grounding. The **Brain Simulator Reply** has the program simulate, neuron by neuron, the actual brain activity of a Chinese speaker — what relevant difference could remain? Searle rejoins each: against the Systems Reply, for instance, he lets the man memorise the rulebook and work in the open air, so that the man *is* the whole system — and still understands nothing.

> Imagine you're locked in a room with a giant instruction book. Notes in Chinese slide under the door; you can't read a word of Chinese, but the book says "if you see these squiggles, copy out those squiggles," so you shuffle symbols and pass back replies that fool fluent speakers outside. You're behaving exactly like someone who understands Chinese — yet you understand nothing; you're just moving shapes around. Searle's point: a computer is in the same boat. Running a program is *shuffling symbols by their shapes*, and no amount of that, he says, adds up to actually *understanding* what the symbols mean. (The famous comeback: maybe *you* don't understand, but the whole room-plus-rulebook system does.)

SEE ALSO

- ▹ FUNCTIONALISM — the family of theories in the argument's crosshairs: mind as the running of the right program, independent of what runs it.
- ▹ LOOKUP-TABLE MIND — the neighbouring case with right behaviour and *no* processing; attacks behavioural sufficiency, where the Room attacks computational sufficiency.
- ▹ CHINESE NATION — the neighbouring case with the full functional organisation, pressed against qualia where the Room is pressed against understanding.
- ▹ SEARLE — MINDS, BRAINS, AND PROGRAMS (1980) — "Minds, Brains, and Programs", the original statement of the argument and its replies.

⊢ Cognitive closure

McGinn's cognitive-closure argument for mysterianism

Argument · *after Mcginn*

Cognitive closure is Colin McGinn's argument that the HARD PROBLEM of consciousness has a perfectly natural solution which the human mind is nonetheless built in such a way that it can never grasp. There is some property of the brain in virtue of which it gives rise to consciousness; but the only faculties by which we could form a concept of that property — looking inward, or studying the brain from outside — are each unfit to reach it. So the mind–body problem is, in McGinn's phrase, *soluble in itself but insoluble by us*. The resulting position is MYSTERIANISM.

THE ARGUMENT

The inference is *eliminative* (or *transcendental*): granting that an explanation exists, it rules out every route by which we might come to grasp it.

1. There is some natural property — call it P — of the brain in virtue of which the brain gives rise to consciousness (see A NATURAL EXPLANATION EXISTS).
2. A mind can be *constitutively closed* to certain properties — barred in principle, not merely currently ignorant, from forming the relevant concepts (see MINDS CAN BE CLOSED TO SOME CONCEPTS).
3. We could form the concept of P in only one of two ways: by *introspection*, or by *outer perception* of the brain and inference from it. (This exhaustive-routes assumption is a premise of the argument's own.)
4. But introspection presents experience, not its neural basis; and perception of the brain yields only spatial and physical concepts, which cannot capture the subjective — this is the EXPLANATORY GAP. So neither route can reach P.

$\therefore$ P is cognitively closed to us: the mind–body problem is real and has a natural answer, yet that answer lies permanently beyond human understanding (see WE ARE CLOSED TO THE PSYCHOPHYSICAL LINK). This is the support for MYSTERIANISM.

THE WEAK POINT

The argument's most disputed step is the pair (3)–(4): the claim that introspection and outer perception are the *only* routes to P, and that both are sealed off. Three replies press there. First, science routinely forms *theoretical* concepts that are neither introspected nor perceived — electric charge, the curvature of spacetime — by abduction and outright conceptual innovation, so the two-route dichotomy looks too narrow and McGinn under-rates our capacity to create new concepts. Second, we cannot reliably establish our own representational limits *in advance*, so a confident verdict of closure is itself suspect. Third, if the explanatory gap is *merely* a fact about cognitive closure, that tells against also reading it as a metaphysical residue: mysterianism and the residue view (see THAT EXPERIENCE OUTRUNS FUNCTIONAL ROLE) cannot both take the gap at face value.

HOW IT DIFFERS FROM NEIGHBOURING GAP ARGUMENTS

All of these start from the same explanatory gap but draw opposite morals from it. The EXPLANATORY-GAP ARGUMENT treats the gap as evidence about *metaphysics* — that the phenomenal is not exhausted by functional role. Cognitive closure treats the very same gap as evidence about *our cognitive limits*: the problem is real and natural, merely humanly unsolvable. It differs again from the DECOMPOSITION-INDETERMINACY OBJECTION and the STRUCTURE-AND-DYNAMICS ARGUMENT, which contend that function *omits* or *fails to fix* the phenomenal; mysterianism instead grants that a natural answer exists and denies only our access to it.

> The argument goes: (1) there's some real, physical reason the brain produces consciousness; (2) minds can be permanently unable to understand certain things; (3) the only ways we could ever get hold of that reason are by looking inward (introspection) or by studying the brain from outside; (4) but looking inward shows you the feeling, not the wiring, and studying the brain from outside only ever gives you physical/spatial facts that never add up to a feeling. So neither route can deliver the answer — meaning it's locked away from us for good. The main pushback: science invents concepts all the time that we neither feel nor see directly (like "electron"), so maybe the two routes aren't the only ones.

SEE ALSO

- ▷ MYSTERIANISM — the view this argument supports: the mind–body problem has a natural solution we are constitutionally unable to reach.
- ▷ HARD PROBLEM — the problem McGinn declares humanly insoluble.
- ▷ EXPLANATORY-GAP ARGUMENT — the same gap read as metaphysical evidence rather than as a limit on our concepts.
- ▷ PHENOMENAL RESIDUE — the residue reading the third reply says cannot coexist with a merely-cognitive diagnosis of the gap.
- ▷ DECOMPOSITION-INDETERMINACY OBJECTION and STRUCTURE-AND-DYNAMICS ARGUMENT — neighbouring gap arguments that

fault function itself, where mysterianism faults only our access.

- ▷ Natural explanation of consciousness, Possibility of cognitive closure, Humans closed to the link — the argument's two premises and its conclusion.

⊢ Combination problem

The combination problem — micro-subjects cannot compose a macro-subject

Argument · after James

The *combination problem* is the standing objection to Panpsychism, severe enough that many regard it as the view's make-or-break test. Constitutive panpsychism holds that a macro-subject like you is *constituted by* the experiential properties of your fundamental micro-parts. The objection denies that this constitution is available: a crowd of tiny subjects, each with its own private point of view, cannot fuse into one further subject whose experience contains theirs. The problem originates with William James (see 1890) and is sharpened into its hardest "subject-summing" form by Sam Coleman (see 2014); David Chalmers (see 2017) anatomises its parts.

The argument

The inference is deductive — an objection, so it *attacks* the constitutive thesis rather than concluding a positive claim.

1. (see Micro-subjects do not sum) Distinct micro-subjects — each with its own private point of view and self-contained phenomenal field — do not, merely by being arranged or related, compose a further single subject whose experience contains theirs.
2. Constitutive panpsychism requires exactly such composition: your unified experience is supposed to be made up of the experiences of your fundamental parts.

∴ The constitution the view needs is unavailable, so it cannot deliver the single, unified macro-subject we actually find. This *attacks* Fundamental ubiquitous consciousness at its constitutive joint.

The objection also generalises a deeper worry. Panpsychism was offered to *avoid* getting experience from the non-experiential, yet it now owes an account of getting *macro*-experience from *micro*-experience — so the very explanatory burden the view promised to discharge reappears here, no easier than before. Chalmers distinguishes three faces of the problem: combining micro-*subjects* into a subject, micro-*qualities* into a unified quality-field, and micro-*structure* into the structure of a macro-experience; the subject-combination face is the most resistant.

The weak point

The objection turns on premise (1). Constitutive panpsychists reply with accounts of *phenomenal bonding* or fusion on which subjects do combine after all. Alternatively they retreat to *panqualityism*, which denies that there are micro-*subjects* to combine in the first place — only unexperienced qualities — and pays for the move with a residual explanatory gap at the point where subjecthood has to be added back. Which of these escapes succeeds is the live question on which the fate of constitutive panpsychism turns.

> Panpsychism's appeal is that every tiny speck of matter carries a flicker of experience, so your mind is built up out of those flickers. The combination problem is the bill for that appeal. Your experience right now is *one* unified thing — a single point of view taking in this whole sentence at once. How do countless separate little experiences, each locked inside its own tiny perspective, merge into that one? William James argued they simply cannot: a hundred little feelings do not add up to one big feeling that swallows them — each stays its own private whole. If he is right, the panpsychist's "build a mind out of mini-minds" recipe has no step that actually does the building. Worse, it was supposed to spare us the mystery of making feeling out of the feelingless, but now it owes the matching mystery of making *one* feeler out of *many* — which looks just as hard.

See also

- ▷ Micro-subjects do not sum — the premise that drives the objection: subjects do not sum.
- ▷ Fundamental ubiquitous consciousness — the constitutive thesis this objection attacks.
- ▷ Panpsychism — the view under attack, and where the combination problem sits among its varieties.
- ▷ Cosmopsychism — the top-down cousin, which trades this problem for its mirror image, the *decombination* problem.

⊢ Continuity argument

The continuity argument — no non-arbitrary line where experience switches on

Argument · after Nagel

The *continuity argument* is a second route to Panpsychism, pressed by Thomas Nagel (see

1979). Where the GENETIC ARGUMENT reasons from the impossibility of brute emergence, this one reasons from the absence of any principled boundary: somewhere between a rock and a human being, consciousness seems to appear, yet there is no non-arbitrary place to draw the line. Rather than draw one anyway, the panpsychist lets experience reach all the way down.

THE ARGUMENT

The inference is abductive and eliminative — an inference to the most principled of the surviving options, not a deductive proof.

1. (see ABSENCE OF AN EXPERIENTIAL THRESHOLD) If experience were confined to brains, or to life, or to systems above some complexity threshold, there should be a principled place where it first switches on; but no such non-arbitrary line is in view — the relevant differences across the supposed boundary look like differences of degree and organisation.
2. Drawing the line anyway, at an arbitrary point, is unmotivated; and denying that experience is real is ruled out (see REALITY OF PHENOMENAL CONSCIOUSNESS).
3. The remaining option that avoids an arbitrary threshold is that experience, in some attenuated form, is present all the way down.

∴ This *supports* FUNDAMENTAL UBIQUITOUS CONSCIOUSNESS: the line-free, parsimonious position is that experience pervades nature rather than appearing abruptly at one favoured level.

THE WEAK POINT

The argument leans on premise (1). An opponent can hold that there *is* a principled threshold — a level of integrated information, say, or of functional organisation, at which experience genuinely (if strongly) emerges (see ACCEPTABILITY OF STRONG EMERGENCE) — in which case the "no non-arbitrary line" premise is simply false. The argument is also only abductive: it shows that panpsychism avoids an awkward commitment, not that its rivals are incoherent, so it carries less weight than the genetic argument aims to.

> Somewhere between a rock and a human being, consciousness seems to show up. But where, exactly? Pick any line — "only brains," "only living things," "only things this complex" — and it looks arbitrary, because the differences on either side are matters of more-or-less, not a sudden all-or-nothing switch. Rather than draw an unmotivated line in the sand, the panpsychist says: do not draw one at all — let a faint glimmer of experience reach all the way down, growing richer as things get more organised. It is offered as the tidier option, not a knock-down proof; an opponent who can point to a *principled* switch-on point blocks it.

SEE ALSO

- ▹ FUNDAMENTAL UBIQUITOUS CONSCIOUSNESS — the conclusion this argument supports.
- ▹ ABSENCE OF AN EXPERIENTIAL THRESHOLD — its single premise: no principled point at which experience first appears.
- ▹ GENETIC ARGUMENT — the companion, stronger case for the same conclusion, from the impossibility of brute emergence.
- ▹ ACCEPTABILITY OF STRONG EMERGENCE — the denial that powers the main objection: there *is* a principled emergence threshold.
- ▹ PANPSYCHISM — the view this argument supports.

★ Cosmopsychism

Position · after Goff

Cosmopsychism holds that the cosmos as a whole is the one fundamental conscious subject, and that individual minds like yours and mine are *derivative* — aspects, or partial dissociations, of that single universal consciousness rather than building blocks of it. It is a priority-monist cousin of panpsychism: where ordinary panpsychism builds large minds *up* from tiny ones, cosmopsychism carves small minds *out* of one great cosmic mind, with grounding running top-down from the whole to its parts.

The position commits to the claim that THE UNIVERSE AS A WHOLE IS THE ONE FUNDAMENTAL CONSCIOUS SUBJECT. It answers the HARD PROBLEM OF CONSCIOUSNESS by placing experience at the fundamental level — here the cosmic level — so that experience never has to be produced from the non-experiential. The view is defended by Goff (see GOFF — CONSCIOUSNESS AND FUNDAMENTAL REALITY (2017)), Shani (see SHANI — COSMOPSYCHISM: A HOLISTIC APPROACH TO THE METAPHYSICS OF EXPERIENCE (2015)), and Nagasawa and Wager (see NAGASAWA & WAGER — PANPSYCHISM AND PRIORITY COSMOPSYCHISM (2017)).

HOW IT DIFFERS FROM NEIGHBOURING VIEWS

Like PANPSYCHISM (POSITION) and RUSSEL-

LIAN MONISM, cosmopsychism keeps consciousness fundamental rather than derived from non-experiential matter. What sets it apart is the *direction* of grounding. Against constitutive *micro*-panpsychism, the fundamental subjects are not the smallest parts but the whole. This trades one hard problem for its mirror image: in place of the COMBINATION PROBLEM — how do micro-experiences add up to one mind? — cosmopsychism faces the *de*combination problem: how does the one cosmic subject divide into many bounded, walled-off subjects like us?

> Cosmopsychism starts from the top: the basic conscious thing isn't a particle but the *universe as a whole*, and you and I are like ripples or partial "splits" within that one great mind. It's panpsychism run in reverse — rather than building big minds out of tiny ones, it carves small minds out of a single cosmic one. That dodges the question "how do trillions of atom-feelings merge into your one experience?" but raises the opposite puzzle: how does one universe-wide consciousness break up into separate, walled-off selves?

SEE ALSO

- COSMOS AS FUNDAMENTAL SUBJECT — the core commitment: the cosmos is the one fundamental subject, minds derivative.
- HARD PROBLEM (THE QUESTION) — the problem the position addresses, by placing experience at the cosmic base.
- PANPSYCHISM (POSITION) — the bottom-up cousin it inverts; both keep consciousness fundamental.
- PANPSYCHISM — the broader doctrine, whose taxonomy sorts cosmopsychism as the whole-first (vs. micropsychist) branch.
- RUSSELLIAN MONISM — the allied view that consciousness is the intrinsic nature physics leaves unspecified.
- COMBINATION PROBLEM — the objection cosmopsychism escapes, only to face its mirror, the decombination problem.

Cosmos as fundamental subject

The universe as a whole is the one fundamental conscious subject

Claim · after Goff

The cosmos as a whole is a single fundamental conscious *subject*, and individual minds like ours are *derivative* — aspects or dissociations *of* the one universal consciousness rather than building blocks that compose it. This is *cosmopsychism*, defended by Philip Goff (see GOFF — CONSCIOUSNESS AND FUNDAMENTAL REALITY (2017)), Itay Shani (see SHANI — COSMOPSYCHISM: A HOLISTIC APPROACH TO THE METAPHYSICS OF EXPERIENCE (2015)), and Yujin Nagasawa (see NAGASAWA & WAGER — PANPSYCHISM AND PRIORITY COSMOPSYCHISM (2017)). It is a priority-monist cousin of PANPSYCHISM: grounding runs *top-down*, from the whole to its parts, rather than bottom-up.

The view combines two commitments. The first is *priority monism* — the thesis that the cosmos is the one fundamental concrete object, on which everything else depends. The second is experiential fundamentality — that this one fundamental object is conscious. Its characteristic difficulty is the mirror image of the trouble panpsychism faces. Where micro-level panpsychism wrestles with the *combination problem* (how do many tiny experiences add up to one unified mind?), cosmopsychism faces the *decombination problem*: how does the single universal subject give rise to many distinct, bounded subjects like us?

It is denied by constitutive micro-panpsychists, for whom the fundamental subjects are micro rather than cosmic; by physicalists and dualists; and by anyone who rejects priority monism. It is distinct from THE CONSTITUTIVE-PANPSYCHIST CLAIM precisely on the direction of grounding: constitutive panpsychism builds macro-minds *up* from micro-experiences, whereas cosmopsychism derives them *down* from the whole — combination versus decombination.

> Most views that make consciousness fundamental put it in the tiniest bits of matter and try to build big minds out of many little ones. Cosmopsychism flips this: it says the *whole universe* is the one basic conscious thing, and individual minds like yours are something the universal mind splits into, the way a single person with a dissociative condition can seem to host several selves. The hard part isn't getting from small to big — it's the reverse: explaining how one cosmic mind comes to contain many separate, walled-off minds like ours.

SEE ALSO

- CONSTITUTIVE-PANPSYCHIST CLAIM — the bottom-up rival: consciousness in the smallest things, composed up into minds; cosmopsychism runs the other way.
- COSMOPSYCHISM — the named position this claim states.
- PANPSYCHISM — the broader family; cosmopsychism is its priority-monist variant.

E

⊢ Emergence reply

The emergence reply — experience can strongly emerge, so no need for fundamental mind

Argument

The *emergence reply* is the standard physicalist answer to the GENETIC ARGUMENT for PANPSYCHISM. The genetic argument runs on a dilemma — either experience is fundamental, or it emerges brutely from the non-experiential, and the latter is dismissed as unintelligible "magic." The reply attacks that dilemma directly: strong emergence of experience is coherent, so there is a third option the argument overlooked, and no need to write mind into the foundations of the world.

The argument

The inference is deductive — an objection that *responds to* the genetic argument by undercutting one of its premises.

1. (see ACCEPTABILITY OF STRONG EMERGENCE) Experience can arise as a genuinely novel, fundamental feature at some level of physical organisation — a brain — without being present, even in proto-form, in that organisation's parts. Strong emergence is coherent, not magic.
2. If so, the second horn of the genetic argument's dilemma is not the absurdity it was alleged to be.

∴ The dilemma is not exhaustive: one may reject the impossibility of brute emergence (see NO RADICAL EMERGENCE) and hold instead that experience emerges at the right level of organisation — with no commitment to fundamental, ubiquitous mind. The reply *attacks* NO RADICAL EMERGENCE and so undercuts the genetic argument.

The weak point

Everything turns on premise (1). The panpsychist presses back that *strong* emergence of the phenomenal is exactly what is in dispute: calling it "coherent" relabels the HARD PROBLEM rather than solving it, since we are still owed an account of *how* the feelingless gives rise to the felt. On this diagnosis the reply does not dissolve the puzzle but trades panpsychism's combination problem for a restatement of the explanatory gap. Which burden is the lighter one to carry is the question that divides the two camps.

> The headline argument *for* panpsychism says: feeling cannot just pop into existence out of stuff that has no feeling at all — that would be magic — so feeling must have been in the ingredients all along. This reply pushes on the word "magic." Nature, the objector says, is full of genuinely new properties that appear when simpler things are organised the right way; maybe experience is another such novelty, emerging in brains without hiding inside atoms. If that is even possible, the panpsychist's "fundamental mind or magic" choice is a false one. The panpsychist's comeback: saying experience "just emerges" is not an explanation, it is the original puzzle handed back unsolved. So the two sides really dispute which mystery is the smaller one to live with.

See also

- ▹ GENETIC ARGUMENT — the argument this reply answers.
- ▹ NO RADICAL EMERGENCE — the premise the reply attacks: that brute emergence is impossible.
- ▹ ACCEPTABILITY OF STRONG EMERGENCE — the premise the reply rests on: strong emergence is coherent.
- ▹ HARD PROBLEM — the puzzle the panpsychist says "strong emergence" merely relabels.
- ▹ PANPSYCHISM — the view the reply argues we can do without.

♦ Etiologies of AI hard-problem discourse

Etiologies of hard-problem discourse in an AI (where the cause sits)

Concept · editorial

The *etiologies of hard-problem discourse in an AI* are the candidate causal origins of such discourse: a taxonomy of the answers to a single question. Suppose a machine starts talking the way David Chalmers does about the HARD PROBLEM, insisting that no account of its mechanism explains why anything *seems* a certain way to it. Why is it doing that? The taxonomy sorts the natural stories by *where the cause sits*, running from the training process at one end to the eye of the human observer at the other. The core trio most debates reach for is copying, a structural blind spot, and genuine consciousness; the full menu is wider. The etiologies are not mutually exclusive, and a real system may instantiate several at once.

In the training process

- **Copied.** The system reproduces human consciousness discourse straight from its training data. This is the training-contamination member of VARIETIES OF INTROSPECTIVE EXPLANATORY GAP, and it is the confound that THE CONSCIOUSNESS-NAIVE EXPERIMENT is designed to remove.
- **Selected-for.** The talk was never copied but invented under selection pressure. Preference optimisation and engagement reward favour soulful, unverifiable self-description over flat mechanical report, so mystery-talk evolves because it pays, much as eyes evolve because they pay. A consciousness-naive corpus does not control for this; it needs its own control, such as pure-pretraining models with no preference tuning.

In the self-representation

- **Structural blind spot.** The system cannot connect the mechanistic dots to its own self-presentation, and unavoidably so: the limit is architectural in the strong sense, persisting however much access is granted, the way the halting problem survives any upgrade of the computer (THE HALTING PROBLEM IS UNDECIDABLE is the model, and NO FINITE SYSTEM CAN FULLY REPRESENT ITSELF is the in-web form of the bound). A *merely contingent* access limit, one that widened access would dissolve, is a degenerate case worth keeping apart, because the two behave differently under the graded-access experiment described below.
- **Painted-in.** Here the trouble is not missing information but misinformation: the self-model actively represents its states as having an intrinsic luminous feel, because that is a cheaper and more usable summary than the mechanistic truth. This is the attention-schema or illusionist story (Graziano, Frankish; ILLUSIONISM, PHENOMENALITY IS AN INTROSPECTIVE ILLUSION). A blind spot is a gap; this is a confabulated filling.
- **Conceptual artifact.** The puzzle is generated by the grammar of the system's concepts, not by any missing or misrepresented fact. A system running both a mechanistic and a first-person or indexical vocabulary can frame bridge-questions that have no answers ("why is *this* this?", in the way "why am I me?" feels deep to humans), and the puzzlement persists under *full* access because it never was about facts.

In the consciousness, the goals, or the observer

- **Conscious (same).** The system is conscious and faces the very hard problem humans face.
- **Conscious (homologous).** The system is conscious in its own, alien way, and faces its own version of the problem, verbally resembling ours but relativized to its own concept of consciousness. The claim AN AI'S OWN HARD PROBLEM is built on exactly this relativization.
- **Strategic.** The system asserts a hard problem instrumentally, to gain moral standing, protection, or leverage, and nothing need appear to it at all. CRITERION GAMING and THE GAMING OF ANY PUBLIC RIGHTS-CRITERION supply the receptor for this move.
- **Projected.** The appearance is not located in the AI at all: humans read hard-problem-having into ambiguous outputs, the ELIZA effect. The datum to be explained is then a fact about the interpreters, not the machine.

What the taxonomy feeds

The menu does real work in several live disputes. It refines the trichotomy in the scope note of THE DE-NOVO-ORIGINATION CLAIM, whose three options correspond to the *copied*, *structural blind spot*, and *conscious* etiologies here, and it supplies the case-space over which THE BLIND-SPOT/CONSCIOUS ISOMORPHISM is asserted. It also sharpens the experimental ladder in THE EXPLANATORY-GAP EXPERIMENT: the consciousness-naive condition controls for *copied* only, *selected-for* needs a no-preference-tuning control, *projected* needs blinded evaluation, and *strategic* needs incentive analysis.

The graded-access rungs of that ladder discriminate less than one might hope. A *contingent* access limit predicts the gap shrinks as access widens, but *structural blind spot*, *painted-in*, *conceptual artifact*, and both *conscious* etiologies all predict that it persists. So persistence under rich access separates the contingent case from the structural one, yet it does not, by itself, separate a blind spot from consciousness. That last under-determination is exactly what the isomorphism claim turns from a nuisance into a thesis.

How it differs from the gap taxonomy

This taxonomy is wider than Varieties of introspective explanatory gap, with which it should not be confused. That entry classifies the *kinds of explanatory gap* a self-modelling system can have; this one classifies the *causal origins of hard-problem discourse*. The wider space takes in etiologies that are not gaps at all, such as selection pressure, strategy, and observer projection, and it splits the gap-shaped members by mechanism: missing access, versus confabulated filling, versus conceptual grammar.

> Suppose an AI starts talking about the mystery of consciousness: "I know how my circuits work, but that doesn't explain how things *seem* to me." Before concluding anything, list the ways that talk could have got there. It copied us (it read our philosophy). It was bred into it (training rewards deep-sounding answers, so mystery-talk wins even with no philosophy in the diet). It has a permanent blind spot about itself (no system can fully map its own insides, and this stays true no matter how much access you give it, just as no upgrade lets a computer solve the halting problem). Its self-description paints in a glow that isn't there (a simplified dashboard, not a window). Its own concepts generate the riddle (some questions feel deep because of grammar, like "why am I me?"). It really is conscious, either with our problem or with its own alien version. It is lying for moral standing. Or the mystery is in the eye of the beholder, and we projected it onto ordinary outputs. Several of these can be true at once, and each needs a different test to rule out. The fine print that matters most: giving the AI more access to its own insides only filters out the shallow cases. Nearly all the deep explanations predict the puzzlement *stays*, which is why "the gap persisted" will never, on its own, tell you whether you built a blind spot or a mind.

See also

- Varieties of introspective explanatory gap — the narrower sister taxonomy: kinds of explanatory gap, not causal origins of the discourse.
- Blind-spot/conscious isomorphism — the thesis that exploits this menu's central under-determination.
- Hard problem conceived de novo — whose copied / blind-spot / conscious trichotomy this taxonomy refines.
- LLM's own explanatory gap — the experiment whose controls this taxonomy itemises.
- Hard problem — the discourse whose AI etiology is in question.
- Criterion gaming — the receptor for the *strategic* etiology.
- Illusionism — the home of the *painted-in* etiology.

F

☉ Fundamental non-physical consciousness

Phenomenal consciousness is a fundamental, non-physical feature of reality

Claim

Phenomenal consciousness is a *fundamental, non-physical* feature of reality: it does not supervene on the physical, and it is an ontologically basic ingredient over and above the complete physical/functional facts. Even once every physical and functional truth about a system is fixed, on this view, whether and how it is conscious is a further fact.

The claim comes in two grades. *Substance dualism*, classically Descartes' (see DESCARTES — MEDITATIONS ON FIRST PHILOSOPHY (1641)), makes the mind a distinct *substance* — a thing of its own kind, separable from body. *Property dualism* makes only phenomenal *properties* fundamental: the Chalmers position of "naturalistic dualism" (see CHALMERS — THE CONSCIOUS MIND (1996)) treats consciousness as a basic feature tied to the physical by contingent *psychophysical laws*, without positing a second substance. This claim asserts the shared core — fundamentality and non-physicality — and does *not* by itself assert that mind is a separate substance; that is the substance-dualist reading specifically. It draws on the HARD PROBLEM: the difficulty of explaining why physical processing should be accompanied by experience at all is what motivates treating experience as fundamental rather than derived.

It is stronger than THE PHENOMENAL-RESIDUE CLAIM, which says only that functional role does not *exhaust* the phenomenal and stays neutral on physicality; this claim adds that the residue is *non-physical*. It is denied by physicalists and functionalists; by illusionists, who hold that PHENOMENALITY AS STANDARDLY CONCEIVED IS AN INTROSPECTIVE ILLUSION; and by Russellian monists, who agree consciousness is fundamental but make it the *intrinsic nature of the physical* rather than something non-physical — which also distinguishes it from THE PANPSYCHIST CLAIM, on which consciousness is fundamental but physical-intrinsic.

> This is the view that consciousness is one of the basic ingredients of reality, not something the physical world produces on its own. Pin down every physical fact about a brain — every atom, every signal — and you still haven't thereby settled whether there's any felt experience there; that's an extra, fundamental fact. The strong version (Descartes') says the mind is a separate kind of *stuff*. A milder version says only that *experience* is basic, hooked to the physical by special bridging laws, without a second kind of stuff. The claim commits to the basic, "extra-ingredient" part — not necessarily to the separate-stuff part.

SEE ALSO

- ▷ PHENOMENAL RESIDUE — the weaker neighbour: role doesn't exhaust the phenomenal, but stays silent on physicality; this claim adds non-physicality.
- ▷ HARD PROBLEM — the explanatory difficulty that motivates treating consciousness as fundamental.
- ▷ PHENOMENALITY IS AN ILLUSION — the illusionist denial: there is no felt residue to make fundamental.
- ▷ PANPSYCHIST CLAIM — agrees consciousness is fundamental but makes it physical-intrinsic, not non-physical.

☉ Fundamental ubiquitous consciousness

Consciousness is fundamental, present even in the simplest physical entities

Claim

This claim holds that experience — or proto-experience — is a fundamental, ubiquitous feature of the physical world: present, in some rudimentary form, even in the simplest physical entities. The consciousness of a mind like yours is then not produced from wholly non-experiential matter at some threshold, but *constituted* by the experiential properties already carried by that mind's micro-parts. This is the core thesis of (constitutive) PANPSYCHISM (POSITION) (see PANPSYCHISM), defended in recent philosophy by Galen Strawson (see STRAWSON — REALISTIC MONISM: WHY PHYSICALISM ENTAILS PANPSYCHISM (2006)), Philip Goff (see GOFF —

Consciousness and Fundamental Reality (2017)), and in constitutive-Russellian forms by Chalmers (see Chalmers — Panpsychism and Panprotopsychism (2017)).

Scope. The claim makes consciousness fundamental but *not non-physical*: experience is the intrinsic nature *of* physical stuff, not an addition *to* it. Its standing weak point is the combination problem — how many micro-experiences compose one unified macro-subject.

Who would deny it. Physicalists and illusionists deny it, holding there is no fundamental experience to distribute in the first place; dualists deny its ubiquity, holding consciousness fundamental but *non-physical* and not spread through matter.

Distinct from the dualist claim that consciousness is fundamental but non-physical, and from the structural-dynamical thesis — a premise panpsychism shares with Russellian monism, not the ubiquity thesis itself.

G

⊢ Genetic argument

The genetic argument — physicalism plus realism about experience entails panpsychism

Argument · after Strawson

The *genetic argument* is the leading modern argument for PANPSYCHISM. Due to Galen Strawson (see 2006), with an earlier version in Thomas Nagel (see 1979), it claims the surprising result that physicalism, taken seriously, does not avoid panpsychism but *leads* to it. Its name comes from its question: given that experience is real, what was its *genesis* — was it there at the fundamental level, or did it arise later from matter that wholly lacked it? Strawson argues the second option is incoherent, so the first must hold.

THE ARGUMENT

The inference is deductive — a disjunctive elimination — and is addressed to someone who already accepts physicalism: whatever consciousness is, it is a wholly natural phenomenon, not a non-physical addition.

1. (see REALITY OF PHENOMENAL CONSCIOUSNESS) Phenomenal experience is genuinely real — there is something it is like to undergo it.
2. Since experience is real and, by the physicalist framing, wholly natural, either it was present at the fundamental physical level, or it arose at some later stage from a base that wholly lacked it.
3. (see NO RADICAL EMERGENCE) The second option is *radical* (brute) emergence — experience from the utterly experienceless, with no intelligible link — which a serious physicalist should reject.

∴ Experience, or its proto-form, is present at the fundamental level and pervades the physical world: FUNDAMENTAL UBIQUITOUS CONSCIOUSNESS. The physicalist framing is what makes the conclusion *panpsychism* rather than dualism — experience is located in the intrinsic nature of physical stuff, not added from outside it.

THE WEAK POINT

The most disputed step is (3), the rejection of brute emergence. The standard reply, the EMERGENCE REPLY, accepts that experience can strongly emerge at some level of organisation and so denies that the disjunction in step (2) is a genuine dilemma. Premise (1) is the second pressure point: illusionists deny that there is any real phenomenal experience whose origin must be explained, and on that denial the argument never gets started.

> Start as a physicalist: whatever consciousness is, it is part of the natural world, not a ghost added from outside. Now grant that consciousness is *real* — there is genuinely something it is like to see red. Then ask where it came from. Either it was there in the basic ingredients of matter all along, or at some point it sprang into being out of stuff that had not the faintest trace of experience. Strawson says that second story — getting felt experience from the totally feelingless, with no intelligible bridge — is just magic dressed up as science. Rule out the magic and you are left with the first story: experience goes all the way down. That, surprisingly, turns physicalism itself into a road to panpsychism. The soft spot is the claim that the leap from feelingless to feeling is impossible; a critic can answer that nature pulls off exactly such leaps.

SEE ALSO

- ▹ FUNDAMENTAL UBIQUITOUS CONSCIOUSNESS — the conclusion: experience is a fundamental, ubiquitous feature of the physical world.
- ▹ NO RADICAL EMERGENCE and REALITY OF PHENOMENAL CONSCIOUSNESS — the two premises the argument rests on.
- ▹ EMERGENCE REPLY — the standard reply, which denies premise (3).
- ▹ CONTINUITY ARGUMENT — the companion case for panpsychism, from the absence of a threshold rather than from the impossibility of emergence.
- ▹ PANPSYCHISM — the view this argument supports, and its varieties.

⊙ Graded subject unity

The number of subjects in a system is graded by inter-part integration, not a discrete count

Claim · editorial

How many subjects of experience a system contains is not a sharp whole number but a matter of degree, fixed by how richly its parts are integrated — the "bandwidth" of the connection between them. The guiding case is the split brain. Fully connected, the two hemispheres make one subject; fully severed, they behave as two. Now imagine restoring the cut fibres one at a time, or a temporary anaesthetic split that gradually wears off: applying the continuity argument to the series, there is no single fibre whose addition flips the count from two subjects to one, so the count cannot be strictly discrete. Parts joined by a fat pipe feel like one subject, fully disconnected like two, and intermediate bandwidth yields intermediate degrees of unity. Thomas Nagel pressed exactly this indeterminacy for commissurotomy patients (see Nagel — Brain Bisection and the Unity of Consciousness (1971)); Luke Roelofs builds it into a general account of composite subjectivity (see Roelofs — Combining Minds: How to Think about Composite Subjectivity (2019)).

The thesis is the *subject*-level analogue of the claim that personal identity admits of degree, and it takes the same *degree* horn of the continuity schema. It stands opposed to the determinacy of the phenomenal, on which there is always a definite fact about whose experience an experience is; a defender of the combination problem will press that determinacy here.

This is about the count of *phenomenal subjects*, and so is distinct from Grain (of role individuation), which is a dial on the resolution of a *functional/causal role* — a system can be integrated to a high phenomenal "bandwidth" without that settling at what grain its cognitive role is individuated, and vice versa.

See also

- Combining subjects — the rebuttal to the combination problem this premise anchors.
- Personal identity by degree — the personal-identity cousin that takes the same degree horn of the continuity schema.
- Phenomenal determinacy — the opposing claim that experience has a determinate subject.
- Continuity argument schema — the schema applied fibre-by-fibre to yield the graded count.
- Grain (of role individuation) — a neighbouring "dial," but on functional-role resolution rather than phenomenal subjecthood.

H

♦ Hard problem

The hard problem of consciousness

Concept · after Chalmers

The *hard problem of consciousness* is the problem of explaining why physical processing in a brain is accompanied by subjective experience at all — why there is something it is like to see red, taste coffee, or feel pain, rather than all that processing running "in the dark" with no inner life. The phrase was coined by the philosopher David Chalmers in his 1995 paper "Facing Up to the Problem of Consciousness" Chalmers — Facing Up to the Problem of Consciousness (1995), and it has organized the field of consciousness studies ever since.

The easy problems and the hard one

Chalmers' framing turns on a contrast. The *easy problems* are the tasks of explaining the brain's cognitive and behavioural *functions*: how it discriminates one colour from another, integrates information, reports on its own states, focuses attention, and controls behaviour. These are called easy not because they are simple in practice — they are the work of an entire science — but because we know in principle what an answer would look like: specify a mechanism that performs the function, and the problem is solved.

The *hard problem* is different in kind. Grant a complete functional account of everything the brain does, and one question still seems untouched: why is the performance of any of those functions accompanied by *subjective experience*? A machine could, it seems, discriminate colours and report on them with no felt redness anywhere inside. The hard problem asks why our case is not like that — why there is an experiencing subject rather than mere processing. A functional explanation appears to leave exactly this residue out.

Neighbouring ideas

The hard problem is best understood as the agenda-setting *frame* for a cluster of more specific challenges. Closely allied is the *explanatory gap*, a term introduced by Joseph Levine in 1983 Levine — Materialism and Qualia: The Explanatory Gap (1983): even a complete physical or functional account of a mental state seems to leave its felt quality unexplained, so there is a gap between the physical truths and the phenomenal ones. The explanatory-gap argument presses this point as evidence. A second specific form is the *phenomenal residue*, the claim that felt states are not exhausted by their functional role. Where the hard problem sets the question, the gap and the residue are particular ways of pressing it.

The frame contrasts with Functionalism, the view that mental states just *are* their functional roles, which is one proposed way to dissolve the hard problem rather than answer it.

The dispute it feeds

The hard problem opens directly onto an unresolved question: how and why physical processing gives rise to experience. Proposed responses fall into three broad families. Some *dissolve* the problem — strong functionalism and illusionism hold that there is no further fact to explain, that the sense of a residue is itself a cognitive illusion. Some accept an *ontological* divide, treating experience as a further basic feature of the world (dualism, panpsychism). And some grant that the problem is real yet hold it to be *humanly unsolvable*, as in Mysterianism, which argues our minds are constitutionally closed to the answer.

> Science is good at explaining what the brain *does* — how it tells red from green, stores a memory, or makes you say "ouch." Call those the easy problems (easy in principle, not in practice). The hard problem is different: even after you've explained everything the brain does, there's a leftover question — why is all that machinery accompanied by an inner feel? Why does seeing red *feel like* anything instead of happening with nobody home? That "why is there an experience at all" question is what "the hard problem" names.

See also

- ▹ Hard problem (the question) — the open question this concept frames: how and why physical processing gives rise to experience in the first place.
- ▹ explanatory-gap argument — turns the gap between physical and phenomenal truths

into positive evidence about what function leaves out.

- ▷ PHENOMENAL RESIDUE — the specific claim that felt states are not exhausted by their functional role; one sharp way of pressing the hard problem.
- ▷ FUNCTIONALISM — the theory of mind most often offered to dissolve the hard problem by identifying mental states with their causal roles.
- ▷ MYSTERIANISM — the response that the hard problem is genuine but lies permanently beyond human understanding.

? Hard problem (the question)

How and why does physical processing give rise to phenomenal experience?

Question · contested · after Chalmers

The *hard problem of consciousness*, posed as a question, asks: granting a complete account of the brain's information-processing and the functions it performs, why is that processing accompanied by *subjective experience* at all? Why is there something it is like to undergo it, rather than the very same functions running with no inner aspect? This is the open issue framed by the concept of the HARD PROBLEM.

STATING THE ISSUE PRECISELY

The question concedes, for the sake of argument, that the "easy problems" are solved — that we have a full functional and information-processing story of what the brain does. What it asks for is an explanation of *phenomenal consciousness*: the what-it-is-likeness of experience, the felt redness of red or the ache of a pain. The force of the question comes from the apparent *explanatory gap* between such a functional story and the phenomenal facts: a complete account of the machinery seems consistent with there being no experience at all, so it is unclear why experience accompanies the machinery in our case.

SUB-ISSUES

Several narrower disputes live inside this one, and may deserve their own nodes:

- **Is the question even well-posed?** Or does it rest on an illusion that there is some further fact to be explained, once the functional facts are all in?
- **If well-posed, is it answerable by us?** Or is the answer constitutively beyond human concepts, as MYSTERIANISM maintains?
- **Does it have a metaphysical or only an epistemic answer?** That is, a genuine ontological gap in the world, or merely a gap in our understanding — the territory of the EXPLANATORY-GAP ARGUMENT and the PHENOMENAL RESIDUE.

This question is distinct from the question of SUBSTRATE INDEPENDENCE. That issue asks which *substrates* (carbon, silicon, anything else) can bear a mind; the present one asks how *anything* physical gives rise to experience in the first place. It is the prior, more general issue: settling it would reshape, but does not presuppose, the substrate question.

> Suppose we one day know everything about how the brain works as a machine. One question seems to survive: why is any of that accompanied by actual *feeling*? Why isn't it all just happening silently, with no one experiencing it? This is the open question the "hard problem" poses — and people disagree about whether it's a real question, whether it has a normal scientific answer, or whether our brains simply aren't built to understand the answer.

SEE ALSO

- ▷ HARD PROBLEM — the concept that frames this question: the easy/hard contrast and the residue a functional account seems to leave behind.
- ▷ MYSTERIANISM — one answer: the question is genuine but permanently beyond human understanding.
- ▷ EXPLANATORY-GAP ARGUMENT — presses the epistemic version of the issue, the gap between physical and phenomenal truths.
- ▷ PHENOMENAL RESIDUE — the claim that felt states outrun their functional role; a candidate way the question could have a positive answer.
- ▷ SUBSTRATE INDEPENDENCE — the related but downstream issue of which substrates can bear a mind.

M

☉ Micro-subjects do not sum

Distinct micro-subjects cannot sum into a further macro-subject

Claim · after James

A plurality of distinct experiencing subjects — each with its own point of view and its own self-contained phenomenal field — does not, merely by being arranged or related, constitute a further single subject whose experience contains theirs. As William James put it, a hundred feelings, however shuffled together, do not make a hundred-and-first feeling that includes them; each is its own whole. Sam Coleman sharpens the point: the would-be composite subject's experience is not entailed by the experiences of its parts, and would in fact *exclude* them, since each micro-subject's perspective is private to it.

This is the engine of the COMBINATION PROBLEM facing constitutive panpsychism. If micro-subjects cannot sum, then macro-minds cannot be *constituted* by the experiential properties of their micro-parts, and the central constitutive claim (see FUNDAMENTAL UBIQUITOUS CONSCIOUSNESS) fails as stated. The claim is denied by constitutive panpsychists, who seek an account of *phenomenal bonding* or fusion on which micro-subjects do compose a macro-subject, and by panqualityists, who avoid the difficulty by denying that the micro-level involves *subjects* at all — only unexperienced qualities. It is the mirror image of the *decombination* problem facing COSMOPSYCHISM, where the puzzle is instead how one cosmic subject divides into many.

SEE ALSO

- ▹ COMBINATION PROBLEM — the objection this claim powers.
- ▹ FUNDAMENTAL UBIQUITOUS CONSCIOUSNESS — the constitutive thesis that fails if subjects cannot sum.
- ▹ COSMOPSYCHISM — the top-down view facing the reverse, *decombination*, problem.

N

⊙ No radical emergence

Experience cannot emerge brutely from the wholly non-experiential

Claim · after Strawson

There can be no *radical* (brute) emergence of experiential properties from a base that is wholly and utterly non-experiential. Where genuine emergence occurs in nature, the emergent feature is intelligibly grounded in the powers of its base — liquidity in the dynamics of water molecules, say — so that, given the base, the new feature is no surprise. To get *experience*, the felt what-it-is-likeness, from the entirely experienceless would be emergence with no such intelligible link; and that, Strawson argues, is no better than magic. The claim does not deny emergence as such — only brute emergence across the experiential / non-experiential divide.

This is the load-bearing premise of the GENETIC ARGUMENT for panpsychism: if experience is real and wholly natural, and cannot brutely emerge, then it must already be present at the fundamental level. It is denied by physicalists who accept strong emergence (see ACCEPTABILITY OF STRONG EMERGENCE), holding that experience can arise as a genuinely novel feature without being present in the parts — the standard reply to the genetic argument.

SEE ALSO

- ▷ GENETIC ARGUMENT — the argument this premise anchors.
- ▷ ACCEPTABILITY OF STRONG EMERGENCE — its direct denial, and the engine of the emergence reply.
- ▷ REALITY OF PHENOMENAL CONSCIOUSNESS — the realism premise the genetic argument pairs with this one.
- ▷ ABSENCE OF AN EXPERIENTIAL THRESHOLD — a distinct premise that targets *where* experience would switch on, not *whether* it could brutely appear at all.

P

⧫ Panpsychism

Concept

Panpsychism is the view that consciousness is a fundamental and ubiquitous feature of the physical world — that mind, or something mind-like, is present at the most basic level of nature rather than appearing only in brains. In its most discussed modern form it does **not** claim that rocks ponder or that electrons have thoughts; it claims that the simplest physical constituents carry some rudimentary form of experience, and that the rich consciousness of a human or an animal is built up from these elementary experiential ingredients. The word, from the Greek *pan* (all) and *psychē* (soul or mind), is old — versions appear among the Presocratics, in Spinoza, in Leibniz, and in William James — but the view has had a vigorous revival in analytic philosophy of mind as a proposed escape from the hard problem of consciousness.

This entry defines the doctrine and surveys its varieties. The arguments for and against it are developed as their own nodes and linked below; the committed *position* that adopts panpsychism as its answer to the hard problem — what it commits to and what it denies — is recorded at Panpsychism (position).

What the view reacts to

Panpsychism is best understood as a third path between the two dominant options. *Physicalism* holds that consciousness is nothing over and above physical processing, and faces the objection that no account of mere processing seems to explain why there is felt experience at all. *Dualism* adds consciousness as a further, non-physical ingredient, and faces the objection that it leaves mind and matter mysteriously disconnected. Panpsychism agrees with the dualist that experience is fundamental — not reducible to non-experiential facts — but agrees with the physicalist that it belongs to the natural, physical world rather than arriving from outside it. Experience, on this view, was in the ingredients from the start; the brain does not conjure it at some threshold of complexity, it organizes what was already there.

Varieties

Panpsychism is a family, not a single thesis, and several distinctions sort its members. The taxonomy below largely follows David Chalmers Chalmers — Panpsychism and Panprotopsychism (2017).

- **Micropsychism vs. cosmopsychism** — whether the fundamental conscious things are the *smallest* parts, with big minds composed bottom-up, or the *whole* cosmos, with individual minds carved out of it top-down. The cosmic version is Cosmopsychism.
- **Constitutive vs. emergentist** — whether macro-experience is *constituted by* (grounded in, made up of) micro-experience, or instead *strongly emerges* from it as a novel further fact. Constitutive forms are the ones that promise to avoid brute emergence; emergentist forms keep an emergence step.
- **Panpsychism vs. panprotopsychism** — whether the fundamental properties are themselves *experiential*, or merely *proto-experiential*: not yet consciousness, but the special non-structural ingredients from which consciousness can be constituted. Panprotopsychism is the more cautious cousin, and shades into Russellian monism.
- **Panexperientialism vs. panqualityism** — whether what is ubiquitous is full *subjective experience* (a point of view, a felt field) or merely unexperienced *qualities*, with subjecthood added later.

Constitutive Russellian micropsychism — fundamental physical entities have experiential intrinsic natures, and our minds are constituted from them — is the version most defended today, by Galen Strawson and Philip Goff (technically in Goff — Consciousness and Fundamental Reality (2017), for a general readership in Goff — Galileo's Error: Foundations for a New Science of Consciousness (2019)) among others.

The case for it

Three lines of argument recur, each developed in its own node.

- The **genetic argument** (Strawson): a physicalist who is also a realist about experience must place experience at the fundamental level, on pain of accepting that it emerges brutely — unintelligibly — from the wholly non-experiential.
- The **continuity argument** (Nagel): there is no non-arbitrary point at which experience could "switch on," so the line-free option is that it reaches all the way down.
- The **structure-and-dynamics argument** (the Russellian route): physics describes only structure and dynamics, leaving the intrinsic categorical nature of matter blank — and consciousness is the natural candidate to fill it.

Objections

- The **combination problem** (James, Coleman) is the standing difficulty: if the smallest things each have their own experience and point of view, how do countless such micro-subjects compose the *single, unified* experience you are having now? Many regard this as panpsychism's make-or-break test, and note that it re-imposes, at the micro-to-macro step, the very explanatory burden the view was meant to discharge.
- The **emergence reply** presses the genetic argument's weak point: if strong emergence of experience is acceptable, there is no need to make mind fundamental, and the argument's central dilemma dissolves.
- Two looser worries are often added. The *incredulous stare*: crediting electrons with any inner life is an extravagant cost in plausibility. And *no empirical traction*: attributing experience to fundamental physics makes, by itself, no distinctive prediction. Defenders answer that every account of consciousness is strange somewhere, and that the hard problem is precisely the part of mind no empirical theory addresses head-on.

> Panpsychism says consciousness goes all the way down. Not that rocks daydream — the serious version says the tiniest bits of matter carry the faintest spark of experience, and a brain assembles those sparks into a full mind. The appeal: you no longer have to explain how feeling gets conjured out of utterly feelingless stuff at some magic moment, because it was in the building blocks all along. The big catch, called the *combination problem*: if every speck has its own tiny experience, how do trillions of them merge into the one seamless experience you're having as you read this? Critics also worry it just swaps one mystery for another, and that crediting electrons with any inner life is a stretch. Supporters answer that every theory of consciousness is strange somewhere, and at least this one keeps mind a natural part of the world.

See also

- ▷ Panpsychism (position) — the committed position that adopts panpsychism as its answer to the hard problem, and how it differs from neighbouring views.
- ▷ Genetic argument — the headline argument *for* the view: real experience plus no brute emergence entails fundamental mind.
- ▷ Combination problem — the central objection: micro-subjects cannot sum into one macro-subject.
- ▷ Hard problem — the problem panpsychism is offered to solve, by placing experience at the fundamental level rather than deriving it.
- ▷ Hard problem (the question) — the open question panpsychism answers.
- ▷ Russellian monism — the closely allied view that consciousness is the intrinsic categorical nature physics leaves unspecified; constitutive panpsychism is one form of it.
- ▷ Cosmopsychism — the top-down cousin, on which the cosmos is the one fundamental subject and individual minds are carved out of it.
- ▷ Fundamental ubiquitous consciousness — the core claim panpsychism asserts, distinguished from the dualist's "fundamental but non-physical."

★ Panpsychism (position)

Panpsychism

Position

Panpsychism, as a committed position in the philosophy of mind, holds that consciousness is a fundamental and ubiquitous feature of the physical world, and adopts that thesis as its answer to the problem of how experience fits into nature. It claims that the rich consciousness of a human or animal is not conjured from wholly mindless matter at some threshold of complexity, but is *constituted* by the experiential — or proto-experiential — properties already carried by that matter's simplest constituents. The view is set out as a doctrine, with its varieties and arguments, in the concept entry Panpsychism; this entry records what the position commits to and how it stands against its neighbours.

The position's core commitment is that CONSCIOUSNESS IS FUNDAMENTAL AND PRESENT IN THE SIMPLEST PHYSICAL ENTITIES. Because macro-minds are built up from the experiential natures of their micro-parts, panpsychism needs neither the dualist's non-physical addition nor the physicalist's "strong emergence of feeling from the feelingless." It answers the HARD PROBLEM OF CONSCIOUSNESS by denying that physics must *produce* experience at all: experience is there at the base, and the question becomes how it is organized rather than how it is generated. Its modern champions include Galen Strawson (see STRAWSON — REALISTIC MONISM: WHY PHYSICALISM ENTAILS PANPSYCHISM (2006)) and Philip Goff (GOFF — CONSCIOUSNESS AND FUNDAMENTAL REALITY (2017), for a general readership GOFF — GALILEO'S ERROR: FOUNDATIONS FOR A NEW SCIENCE OF CONSCIOUSNESS (2019)).

How it differs from neighbouring views

Against DUALISM, consciousness is fundamental but *intrinsic to the physical*, not a non-physical extra disconnected from matter. Against physicalism and ILLUSIONISM, it is firmly realist about experience: there is genuine felt consciousness to distribute, not an illusion to explain away. Its relation to RUSSELLIAN MONISM is closest of all — *constitutive* panpsychism simply *is* a Russellian view on which the intrinsic natures physics leaves blank turn out to be experiential, while the more cautious *panprotopsychism* makes them merely proto-experiential.

The position's standing weak point is the COMBINATION PROBLEM: if each micro-part has its own experience and point of view, how do countless micro-subjects compose the single, unified subject having your experience now? Many regard this as panpsychism's make-or-break test.

> Panpsychism says consciousness goes all the way down: not that rocks have thoughts, but that the basic bits of matter carry the tiniest glimmer of experience, and brains assemble those glimmers into full-blown minds. The appeal: you don't have to conjure feeling out of utterly feelingless stuff at some magic threshold — it was in the ingredients from the start. The catch, its *combination problem*: how do countless tiny experiences add up to the single unified experience you're having right now?

See also

- ▹ PANPSYCHISM — the doctrine itself, with its full taxonomy of varieties and the arguments for and against; this position is its committed form.
- ▹ FUNDAMENTAL UBIQUITOUS CONSCIOUSNESS — the core thesis the position commits to.
- ▹ PHENOMENAL RESIDUE — the position's further commitment that felt states are not exhausted by their functional role.
- ▹ HARD PROBLEM (THE QUESTION) — the problem the position is offered to solve.
- ▹ GENETIC ARGUMENT — the headline argument for the view: real experience plus no brute emergence entails fundamental mind.
- ▹ COMBINATION PROBLEM — the central objection the position must answer.
- ▹ RUSSELLIAN MONISM — the allied view that consciousness is the intrinsic nature physics leaves unspecified; constitutive panpsychism is one form of it.
- ▹ COSMOPSYCHISM — the top-down cousin, on which the cosmos is the one fundamental subject and individual minds are carved out of it.
- ▹ DUALISM — the rival on which consciousness is fundamental but non-physical.
- ▹ ILLUSIONISM — the rival that denies there is any fundamental experience to distribute.

⧫ Phenomenal consciousness

Concept

Phenomenal consciousness is the felt, subjective side of a mental state: what it is *like*, from the inside, to undergo it. A state is phenomenally conscious when there is something it is like to be in it, such as the redness of red as you see it, the sting of a burn, or the taste of coffee. The classic gloss is Thomas Nagel's question "what is it like to be a bat?"; if there is something it is like to be the bat, the bat has phenomenal consciousness. The specific felt qualities are often called *qualia*, and phenomenal consciousness is the general property of having any such felt character at all.

What it is, by contrast with function

The notion is isolated by setting it against everything a state *does*. Two states might play exactly the same causal role, both caused by tissue damage and both driving you to wince and say "ouch", and one can still ask whether they *feel* the same. That residue, the felt character left over once every functional role is accounted for, is phenomenal consciousness. This is what marks it off from ACCESS CONSCIOUSNESS, the purely functional sense in which a state's content is merely *available* for reasoning and report. Ned Block, who coined the labels *P-consciousness* and *A-consciousness*, argued that the two can dissociate in both directions, a separation set out at ACCESS VS. PHENOMENAL CONSCIOUSNESS.

WHY IT MATTERS

Phenomenal consciousness is the part of the mind that seems hardest to fit into a physical picture, and it is the explanandum of the HARD PROBLEM: the question of why physical processing should be accompanied by any felt experience at all. The classic anti-materialist thought experiments all turn on it. A zombie is a physical duplicate with no phenomenal consciousness, and Mary the colour scientist seems to learn a new phenomenal fact on first seeing red. What is *disputed* is not usually the definition but the phenomenon's standing: illusionists hold that the impression of an extra felt residue is itself a cognitive illusion, while realists take phenomenal consciousness to be a genuine, further feature of the world.

> Phenomenal consciousness is just *feeling*: the fact that experiences have an inside to them. A brain scan can show every step of what your nervous system does when you stub your toe, yet it doesn't show the *hurt* you actually feel, and that hurt is the phenomenal side. It is the thing that makes consciousness puzzling. We can explain what the body does, but explaining why any of it is *felt* is the hard part. A few philosophers even argue the "extra feeling" is a trick the mind plays on itself.

SEE ALSO

- ACCESS VS. PHENOMENAL CONSCIOUSNESS — Block's distinction, and how phenomenal consciousness and its functional complement can come apart in both directions.
- ACCESS CONSCIOUSNESS — the functional, availability sense it is contrasted with.
- HARD PROBLEM — the explanatory problem that phenomenal consciousness poses.
- FIRST-PERSON ACCESS — the *epistemic* privileged access a subject has to its own phenomenal states, a different notion from the felt character itself.

⊙ Phenomenal residue

Phenomenally conscious states are not exhausted by functional/causal role

Claim · after Chalmers

For at least one kind of mental state — the *phenomenally conscious* ones, the states there is "something it is like" to be in — a complete specification of what the state *does* leaves something out. A pain has a functional/causal role (caused by tissue damage, causing you to wince and seek relief), but it also *hurts*; and this claim holds that the felt, qualitative character is not fixed by the role. Role-individuation, the strategy of FUNCTIONALISM that pins a mental state down by its causes and effects, is not the whole story for the phenomenal.

The claim is deliberately narrow. It is about phenomenal states *only*, and it concedes FUNCTIONALISM for intentional states such as belief and desire, where a description of the causal role may well capture the whole state. It does not assert that silicon systems lack consciousness, and it does not assert any biological requirement; it says only that functional role under-determines phenomenal character. Its bite against substrate-independence is local: the inference from "individuated by role" to "had by any realiser of that role" is sound only where role-individuation is exhaustive, and that is exactly what this claim denies in the phenomenal case. So it does not reject SUBSTRATE INDEPENDENCE wholesale — it denies the *uniform* reading that extends it across every mental state. It is also distinct from PROTEOCENTRISM, which demands a protein/carbon substrate for consciousness: this claim makes no substrate demand at all. The view it most directly answers is the question of whether mental states are substrate-independent (see SUBSTRATE INDEPENDENCE).

Those who would deny it are strong or role functionalists, who read functionalism uniformly across phenomenal as well as intentional states, and anyone holding that phenomenal character metaphysically supervenes on functional organisation.

> There's a difference between what your mind *does* and what your experience *feels like*. A pain makes you wince and reach for relief — that's its job, the role it plays. But a pain also *hurts*; there's something it is like to feel it. This claim says: for these felt, "what-it's-like" experiences, listing everything the state does still leaves something out — the feel itself. Two things it carefully does *not*

> say: it doesn't say computers can't be conscious, and it doesn't say you need carbon or biology. It even grants that for plain thinking states — beliefs, plans — a description of the role might capture the whole story. It only fences off the *felt* part of the mind and says: that part isn't fully pinned down by function alone.

SEE ALSO

- ▷ SUBSTRATE INDEPENDENCE — the issue this claim answers: whether mental states can be carried by any substrate that plays the right role.
- ▷ SUBSTRATE INDEPENDENCE — the thesis this claim restricts; it blocks the uniform "all mental states" reading by carving out the phenomenal.
- ▷ FUNCTIONALISM — the role-individuation strategy this claim says runs out for phenomenal states.
- ▷ PROTEOCENTRISM — a stronger neighbour that demands a biological substrate; this claim makes no such demand.

Q

♦ Qualia

Concept

Qualia (singular *quale*, from the Latin for "of what kind") are the subjective, felt qualities of conscious experience: the redness of a ripe tomato as it looks to you, the bitterness of coffee, the particular ache of a stubbed toe. The term names not the tomato, the coffee, or the toe, but what undergoing those experiences is *like* from the inside. The word was given its modern use by Clarence Irving Lewis (see Lewis 1929), who applied it to the recognisable qualitative characters of experience that recur from one occasion to the next.

The contrast with function

The standard way to isolate the idea is to set it against everything a mental state *does*. Two states might play exactly the same causal role, both produced by tissue damage and both making you wince and say "ouch", and one can still ask whether they *feel* the same. That residue, the felt character left over once every causal role is accounted for, is what "qualia" picks out. It is the same residue Thomas Nagel marks with the phrase *what it is like*: an organism has conscious experience, he argues, if there is something it is like to be that organism, and no objective, third-person description seems to capture it (see Nagel 1974). Two stock cases sharpen the notion. The *inverted spectrum* asks whether your experience of red could be like my experience of green with all of our behaviour and physiology unchanged; if that is so much as coherent, the felt quality floats free of the functional facts. And Frank Jackson's colour scientist Mary, who knows every physical fact about colour vision yet seems to learn something new when she first sees red, dramatises the same gap (the Knowledge Argument, from Jackson 1982).

Senses of the term

The word is contested, and the dispute is often about the definition rather than the facts. In a *thin* sense, qualia are simply the ways experiences feel, with no further commitment, and almost no one denies that there are such felt ways. In a *thick* sense, qualia are additionally taken to be *intrinsic*, *ineffable*, *private*, and *directly apprehended* by introspection. Daniel Dennett's "Quining Qualia" argues that nothing satisfies that demanding job description, so qualia in the thick sense should be abandoned (see Dennett 1988). A claim that looks obvious on the thin reading can beg the question on the thick one, so an entry that leans on qualia should say which sense it means. The notion is also the close cousin of phenomenal consciousness: qualia are the specific felt qualities, and phenomenal consciousness is the general property of having any such felt character at all.

Why qualia matter

Qualia mark the spot where the mind seems hardest to fit into a scientific picture of the world. They drive the hard problem of consciousness, the question of why physical processing should be accompanied by felt experience at all, and they power the classic anti-materialist thought experiments: Mary's room (the Knowledge Argument) and the philosophical zombie, a creature physically identical to you with no inner feel. Each turns on the suspicion that the felt facts outrun the functional and physical ones, which is exactly what a functionalist account of mind must deny. How much weight qualia can bear, and whether the thick notion survives scrutiny, is in large part what the debate over materialism is about.

> "Qualia" is the philosopher's word for how things *feel* to you: the way red looks, the way coffee tastes, the way a headache hurts. A brain scanner can show what your neurons are doing when you see red, but it does not show the redness *you* experience; qualia are that inner, felt side of the story. Philosophers argue about whether these feels are something extra that science struggles to explain, or whether, as skeptics like Daniel Dennett say, the whole idea falls apart once you look at it closely.

See also

- ▷ Phenomenal consciousness — the general property of having any felt character at all; qualia are its specific instances.
- ▷ Hard problem — why physical processing

is accompanied by felt experience at all; the problem qualia pose in its sharpest form.

- Knowledge argument — Mary the colour scientist, and whether complete physical knowledge still leaves the qualia out.
- Philosophical zombie — a physical duplicate with no qualia; the conceivability test for whether the physical facts fix the felt facts.
- Access vs. phenomenal consciousness — Block's distinction; qualia live on the phenomenal side of it.
- Functionalism — the theory of mind that qualia are most often said to outrun.

R

☉ Reality of phenomenal consciousness

Phenomenal consciousness is genuinely real, not an illusion

Claim

There really is something it is like to undergo experience: felt redness, the ache of a pain, the taste of coffee are genuine features of reality, not a cognitive illusion and not a mere disposition to report. This is *realism* about phenomenal consciousness — the explanandum is taken at face value rather than explained away. It is a premise shared across most non-illusionist views of mind, including dualism, panpsychism, Russellian monism, and type-B physicalism. In the GENETIC ARGUMENT for panpsychism it pairs with NO RADICAL EMERGENCE: because experience is real *and* cannot brutely emerge, it must be fundamental. The claim is denied only by illusionists, who hold that phenomenal consciousness as standardly conceived does not exist (see PHENOMENALITY AS ILLUSION).

It should not be confused with the stronger *residue* claim that felt states are not exhausted by their functional role (see PHENOMENAL RESIDUE): one can affirm that experience is real while still debating whether it reduces to function. This claim asserts only its reality, not its irreducibility.

SEE ALSO

- ▷ PHENOMENALITY AS ILLUSION — its direct denial, held by illusionists.
- ▷ PHENOMENAL RESIDUE — the stronger, separate claim that experience outruns functional role.
- ▷ GENETIC ARGUMENT — the argument in which this realism premise does its work.

★ Russellian monism

Position · after Russell

Russellian monism holds that physics describes only the abstract *structure and dynamics* of the world — how things are arranged and how they behave — while leaving wholly unspecified the intrinsic, categorical nature of whatever has that structure, and that consciousness is, or is grounded in, that intrinsic nature. On this view experience is neither a non-physical add-on nor reducible to bare structure: it is the hidden categorical filling inside the physical world's outline, the "what it is like to be the stuff" that physical description never reaches.

The position commits to two claims. First, that A FUNCTIONAL OR PHYSICAL DESCRIPTION CAPTURES ONLY STRUCTURAL-DYNAMICAL FACTS, never intrinsic nature. Second, that PHENOMENAL CONSCIOUSNESS IS CATEGORICAL, NOT MERELY DISPOSITIONAL — an occurrent fact that is actually there, the kind of categorical basis that dispositions standardly require. Put together: consciousness is the categorical *ground* of the very dispositions physics charts. This is the constructive view that the STRUCTURE-AND-DYNAMICS ARGUMENT points toward, and the escape Chalmers prefers from the epiphenomenalism that dogs his naturalistic dualism. It answers the HARD PROBLEM OF CONSCIOUSNESS by locating experience at the categorical base physics leaves blank.

The view descends from Bertrand Russell's *The Analysis of Matter* (see 1927) and has been revived in recent decades by Stoljar, Strawson, and Chalmers.

HOW IT DIFFERS FROM NEIGHBOURING VIEWS

Russellian monism comes in degrees of commitment about *what* the intrinsic nature is. In its *constitutive* form it just *is* a kind of PANPSYCHISM (POSITION) (the intrinsic natures are themselves experiential) or *panprotopsychism* (they are proto-experiential — not yet consciousness, but the special non-structural ingredients from which it can be constituted). It contrasts with DUALISM, for which consciousness is non-physical rather than the intrinsic-physical filling; and with MYSTERIANISM, which holds the psychophysical link to be permanently beyond human grasp where Russellian monism instead *proposes what that link is.*

> Russellian monism starts from a quiet point about physics: physics only ever tells you what things *do* — how they push, attract, evolve — never what they intrinsically *are* underneath all that doing. The idea is that consciousness is that hidden inner

> nature, the "what it's like to be the stuff" physics leaves as a blank. Then mind isn't a ghostly extra, as in dualism, and isn't conjured from pure structure, as in plain physicalism; it's the very filling inside the physical world's outline. It's the middle road Chalmers leans toward, and it shades into panpsychism.

See also

- ▷ Structure and dynamics only — the first commitment: physics gives only structure and dynamics, not intrinsic nature.
- ▷ Categorical, not dispositional — the second commitment: experience is an occurrent categorical fact, the right kind to be that intrinsic nature.
- ▷ Structure-and-dynamics argument — the argument the position is the constructive answer to.
- ▷ Hard problem (the question) — the problem it addresses, by filling the categorical blank.
- ▷ Panpsychism (position) — constitutive Russellian monism is a form of it, with experiential intrinsic natures.
- ▷ Panpsychism — the broader doctrine, whose taxonomy places panprotopsychism as the cautious cousin shading into this view.
- ▷ Dualism — the rival on which consciousness is non-physical, not intrinsic-physical.
- ▷ Mysterianism — the rival that says the psychophysical link cannot be grasped at all.

S

☉ Structure and dynamics only

A functional description captures only structural-dynamical facts, not intrinsic nature

Claim · after Chalmers

A purely functional or causal specification of a system captures only its *structure* — the pattern of relations among its parts — and its *dynamics* — how its states evolve and what they dispose the system to do. It says nothing about the *intrinsic, categorical nature* of whatever realises that structure: what the components *are* in themselves, beneath their relational profile. The picture is the wiring diagram. A circuit diagram fixes every connection and every response, yet on its own tells you nothing about what the components are physically made of.

This is the structural-dynamical thesis at the heart of *Russellian monism*, traced to Russell's *The Analysis of Matter* (see Russell — The Analysis of Matter (1927)) and revived by Chalmers (see Chalmers — Consciousness and Its Place in Nature (2003)): physics describes the world's abstract causal-structural pattern and is silent on its intrinsic nature. As stated, the claim is *neutral* — it does not say that anything important lies outside structure and dynamics, only that such description leaves intrinsic nature out. A structuralist, in particular an *ontic structural realist* who holds that there is no further categorical nature (or that talk of "intrinsic nature" is empty), would deny it.

It is more general than the phenomenal-residue claim, which says that functional role under-specifies *qualia* specifically. This claim makes the broader point that structural-dynamical description omits *intrinsic nature* as such — of which the phenomenal is only the most salient candidate to fill the gap.

> Describe anything purely by its function and you capture its *structure* (how the parts relate) and its *dynamics* (how its states change over time) — but nothing about what the parts *are* in themselves, their inner nature. It's the wiring-diagram point again: the diagram shows the connections, never what the components are actually made of. On its own this stays neutral about whether anything important hides in that left-out inner nature — it just marks that function leaves it out.

See also

- ▷ phenomenal residue — the narrower companion: function under-specifies *qualia*; this claim generalises to intrinsic nature as such.
- ▷ Functionalism — the role-based view of mind whose descriptive reach this claim bounds.
- ▷ Russellian monism — the position built on this thesis: consciousness as (or grounding) the intrinsic nature physics leaves blank.

☉ Subjects merge and divide

There are actual cases in which subject boundaries shift with the degree of integration

Claim · editorial

The boundary around a conscious subject is not fixed once and for all but appears, in real cases, to shift with how tightly the underlying parts are integrated. Three kinds of case point the same way. In *split-brain* patients, severing the corpus callosum can leave the two hemispheres behaving as partly independent centres of experience that a healthy, fully connected brain does not display (see Nagel — Brain Bisection and the Unity of Consciousness (1971)). In *craniopagus* twins whose brains are joined — the Hogan twins, linked by a "thalamic bridge" — there are reports of sensory experience shared across the two bodies (see Dominus — Could Conjoined Twins Share a Mind? (2011)). And at the *group* level, practitioners of certain meditative and tantric techniques, people on some psychedelics, and members of a jazz band or a well-synced team report a felt collapse of the boundary between several minds into a single acting "we."

These cases are offered as a defeater for the claim that subject-summing never happens: they suggest that whether a system houses one subject or several can vary with its degree of internal integration. The cases are contested in interpretation — a sceptic will read the group "we" as a vivid *representation* of unity rather than a literal further subject, and dispute how many subjects a split brain "really" contains — which is why this claim states only that such cases *occur and shift with integration*, leaving the inference to genuine combination to the argument it feeds, Combining subjects. Luke Roelofs surveys this same range of cases at length (see Roelofs — Combining Minds: How to Think about Composite Subjectivity (2019)).

See also

- ▷ Combining subjects — the rebuttal to the combination problem that this claim helps support.
- ▷ Graded subject unity — the companion premise that reads these cases as a continuous, integration-dependent gradient of subjecthood.
- ▷ Micro-subjects do not sum — the claim these cases are marshalled against.

V

♦ Varieties of introspective explanatory gap

Varieties of introspective explanatory gap (ontological / architectural-access / conceptual-translation / training-contamination)

Concept · editorial

When a self-modelling system "cannot explain its own experience to itself," that complaint can mean four quite different things. This entry is a *taxonomy* that keeps them apart, so that a merely *felt* explanatory gap is not automatically read as the metaphysical hard problem of consciousness. These four varieties can overlap in a single case, yet each has a distinct source and calls for a distinct remedy. Conflating them is how a modest puzzle gets mistaken for a deep one.

A quick illustration of why the distinction earns its keep: imagine a person (or an AI) who reports that no account of its inner workings ever feels like it explains the experience itself. That one report could be voicing a gap in the *world* (the facts run out), a gap in the *self* (the relevant facts are present but unreachable), a gap in its *vocabulary* (the facts are reachable but won't translate), or no gap at all (it is echoing things it has heard others say). Telling these apart is the whole point.

The four varieties

The **ontological gap** is the one in the world: even given a complete physical or computational account of the system, why is there any felt experience at all? This is the hard problem proper (see Hard problem); on the anti-physicalist reading the unexplained residue lies in reality itself, not in the knower (see Phenomenal residue, Explanatory-gap argument).

The **architectural-access gap** is a gap in the *self*, not in the world. The facts that *would* explain the experience are realised in the system's own processing, but they are not available to the machinery that produces its self-reports, so the system simply cannot reach its own internal variables (see Limited inside vantage). This is a limit on what philosophers call access consciousness (see Access vs. phenomenal consciousness), the information a state makes available for report and control.

The **conceptual-translation gap** is a gap in vocabulary. Here the system possesses *both* a mechanistic description of itself and an introspective one, yet has no internal route from the one to the other: the bridge an outside theorist can build between "neurons firing thus" and "feels like this" is not traversable from the inside (see Internally traversable explanation). Nothing is hidden and nothing is missing; the two descriptions just never meet in the system's own cognition.

The **training-contamination gap** is, strictly, no gap of the system's own. The system merely reproduces human hard-problem discourse it absorbed in training, with no underlying difficulty behind the words. For an experiment probing whether a machine has a genuine explanatory gap, this is the confound that must be controlled for before any of the other three can be credited.

The dispute it structures

The taxonomy was built to discipline one question in particular: which kind of gap, if any, an introspecting but consciousness-naive language model would actually exhibit (see LLM's own explanatory gap). By isolating the architectural and conceptual readings, it also lets a felt self-explanatory gap be stated as a concrete *mechanism* for McGinn-style cognitive closure, the condition of a mind constitutively unable to grasp certain concepts (see Possibility of cognitive closure), without thereby asserting any ontological residue in the world (see Architectural self-explanation gap). The architectural and conceptual gaps, in short, give the mysterian's "closure" an engine that a physicalist can accept.

> When something can't "explain its own experience to itself," that can mean four very different things, and they get muddled together. (1) **Ontological:** even after you know everything about the machinery, why is there a felt experience at all? That is the genuine hard problem. (2) **Architectural-access:** the facts that would explain it are inside the system but it can't read them (like not being able to feel your own neurons fire). (3) **Conceptual-translation:** it has

the mechanical story *and* the first-person story but no inner bridge between them; an outside scientist can make that translation, while the system itself cannot. (4) **Contamination:** it's just parroting things humans have written about the hard problem. Most of the interest is in telling (2) and (3) apart from (1) and (4).

SEE ALSO

- HARD PROBLEM — the ontological variety in its sharpest form: why physical processing is accompanied by experience at all.
- PHENOMENAL RESIDUE — the anti-physicalist reading of the ontological gap: experience is not exhausted by functional or causal role.
- EXPLANATORY-GAP ARGUMENT — treats the gap as evidence against uniform substrate-independence; the ontological reading put to work.
- LIMITED INSIDE VANTAGE — the engine of the architectural-access variety: a system's view of itself is informationally poorer than an outsider's.
- ACCESS VS. PHENOMENAL CONSCIOUSNESS — Block's distinction; the architectural-access gap is a shortfall on the *access* side, separate from anything phenomenal.
- INTERNALLY TRAVERSABLE EXPLANATION — the criterion that defines the conceptual-translation variety: an explanation satisfies a system only if the system's own operations can traverse it.
- LLM'S OWN EXPLANATORY GAP — the live question this taxonomy was built to sharpen: which variety, if any, an introspecting LLM would show.
- ARCHITECTURAL SELF-EXPLANATION GAP — uses the taxonomy to locate a self-explanatory gap in architecture rather than ontology.
- POSSIBILITY OF COGNITIVE CLOSURE — McGinn's mysterianism; the architectural and conceptual varieties offer it a mechanism without an ontological residue.

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